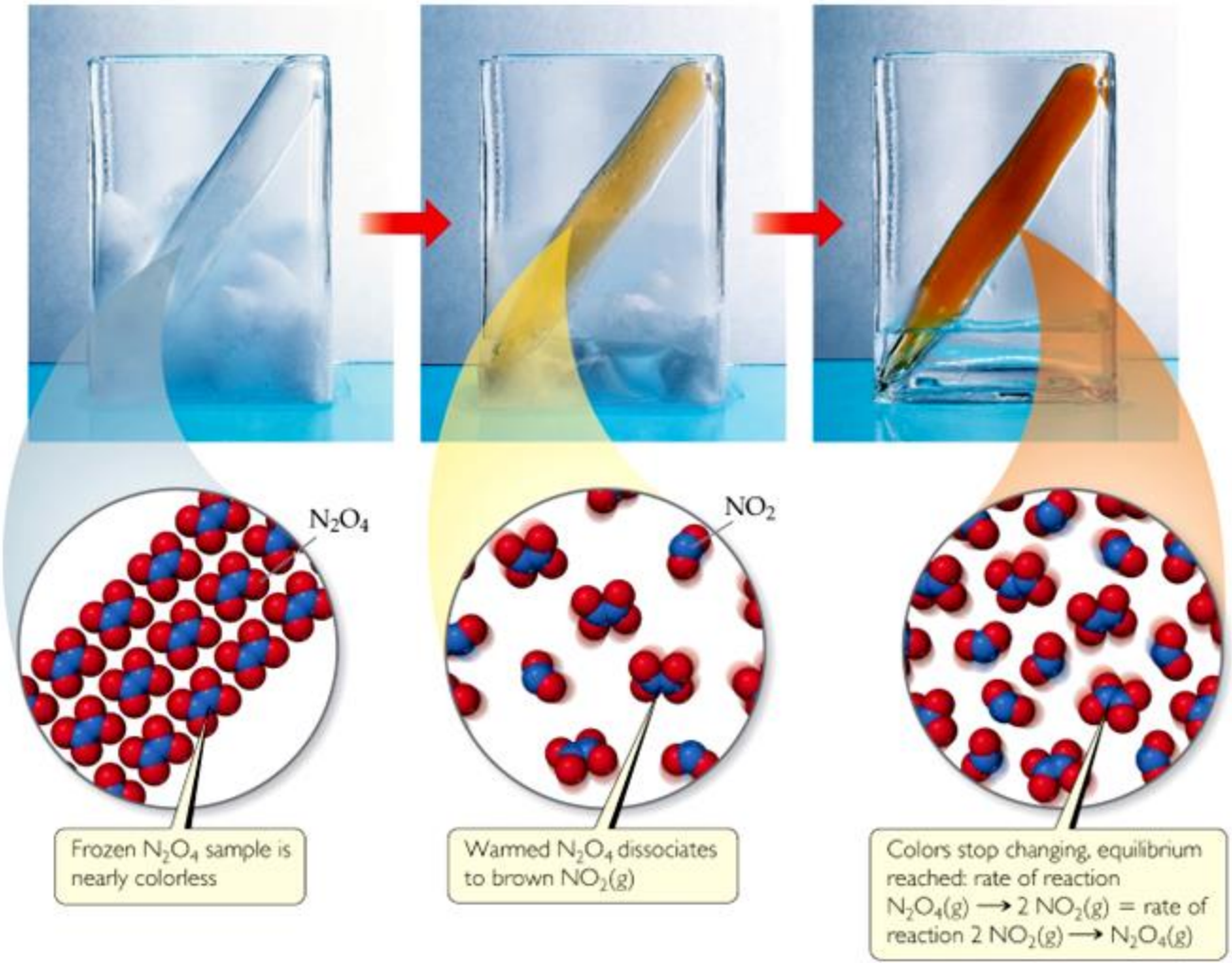
Chapter 15

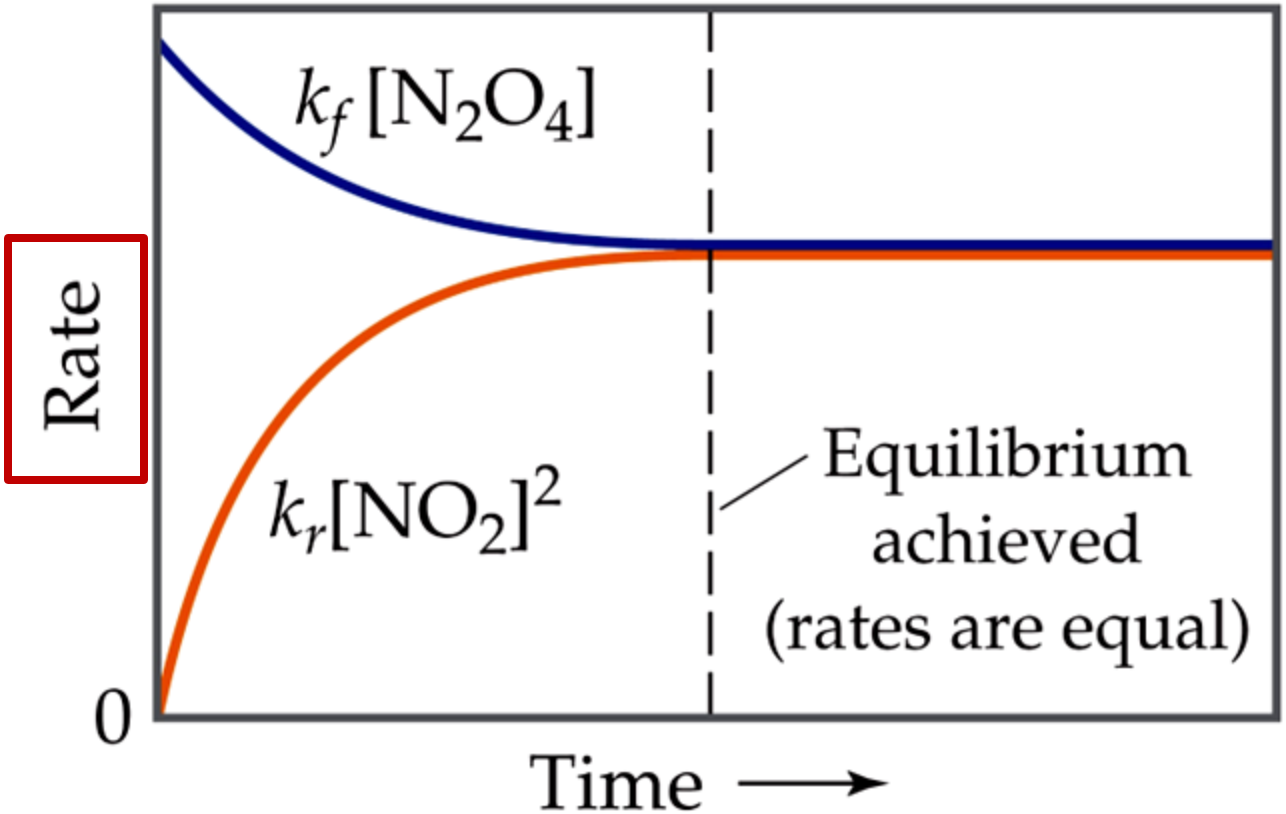
Chemical Equilibrium

The Concept of Equilibrium



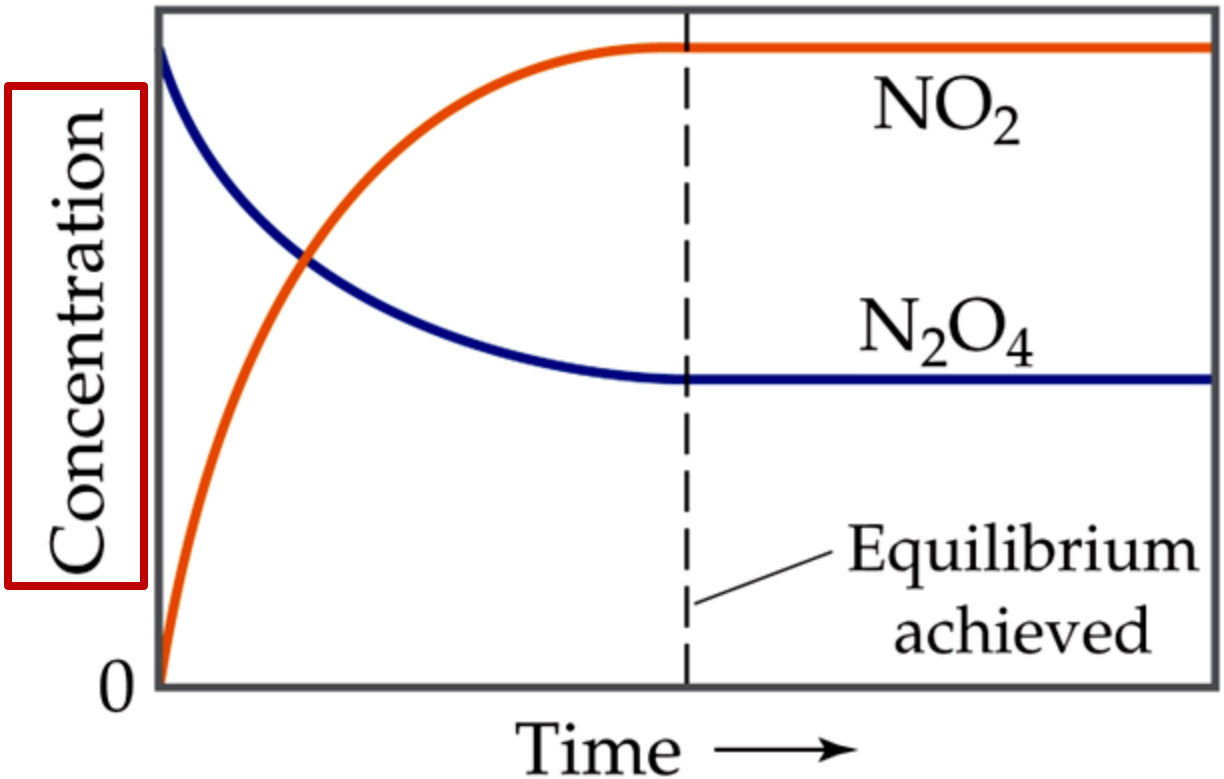
Chemical equilibrium occurs when a reaction and its reverse reaction proceed at the same rate.

The Concept of Equilibrium



- As a system approaches equilibrium, both the forward and reverse reactions are occurring.
- At equilibrium, the forward and reverse reactions are proceeding *at the same rate*.

A System at Equilibrium



Once equilibrium is achieved, the *amount* of each reactant and product remains constant.

Depicting Equilibrium

Since, in a system at equilibrium, both the forward and reverse reactions are being carried out, we write its equation with a double arrow:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The Equilibrium Constant

平衡常数

The Equilibrium Constant

- Forward reaction:

$$N_2O_4(g) \longrightarrow 2NO_2(g)$$

- Rate law:

$$\text{Rate} = k_f[N_2O_4]$$

The Equilibrium Constant

- Reverse reaction:

$$2\ NO_2(g) \longrightarrow N_2O_4(g)$$

- Rate law:

$$\text{Rate} = k_r[NO_2]^2$$

The Equilibrium Constant

- Therefore, at equilibrium

$$\text{Rate}_f = \text{Rate}_r$$

$$k_f[N_2O_4] = k_r[NO_2]^2$$

- Rewriting this, it becomes

$$\frac{k_f}{k_r} = \frac{[NO_2]^2}{[N_2O_4]}$$

The Equilibrium Constant

The ratio of the rate constants is a constant at that temperature, and the expression becomes

$$K_{eq} = \frac{k_f}{k_r} = \frac{[NO_2]^2}{[N_2O_4]}$$

The Equilibrium Constant

- Consider the generalized reaction

$$aA + bB \rightleftharpoons cC + dD$$

- The equilibrium expression for this reaction would be

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

c for concentration expressed in molarity

The Equilibrium Constant

Since pressure is proportional to concentration for gases in a closed system, the equilibrium expression can also be written

$$K_p = \frac{(P_C)^c\,(P_D)^d}{(P_A)^a\,(P_B)^b}$$

p for partial pressure expressed in atm

Relationship between K_c and K_p

- From the ideal-gas law we know that

$$PV = nRT$$

- Rearranging it, we get

$$P = \frac{n}{V} RT$$

Relationship Between K_c and K_p

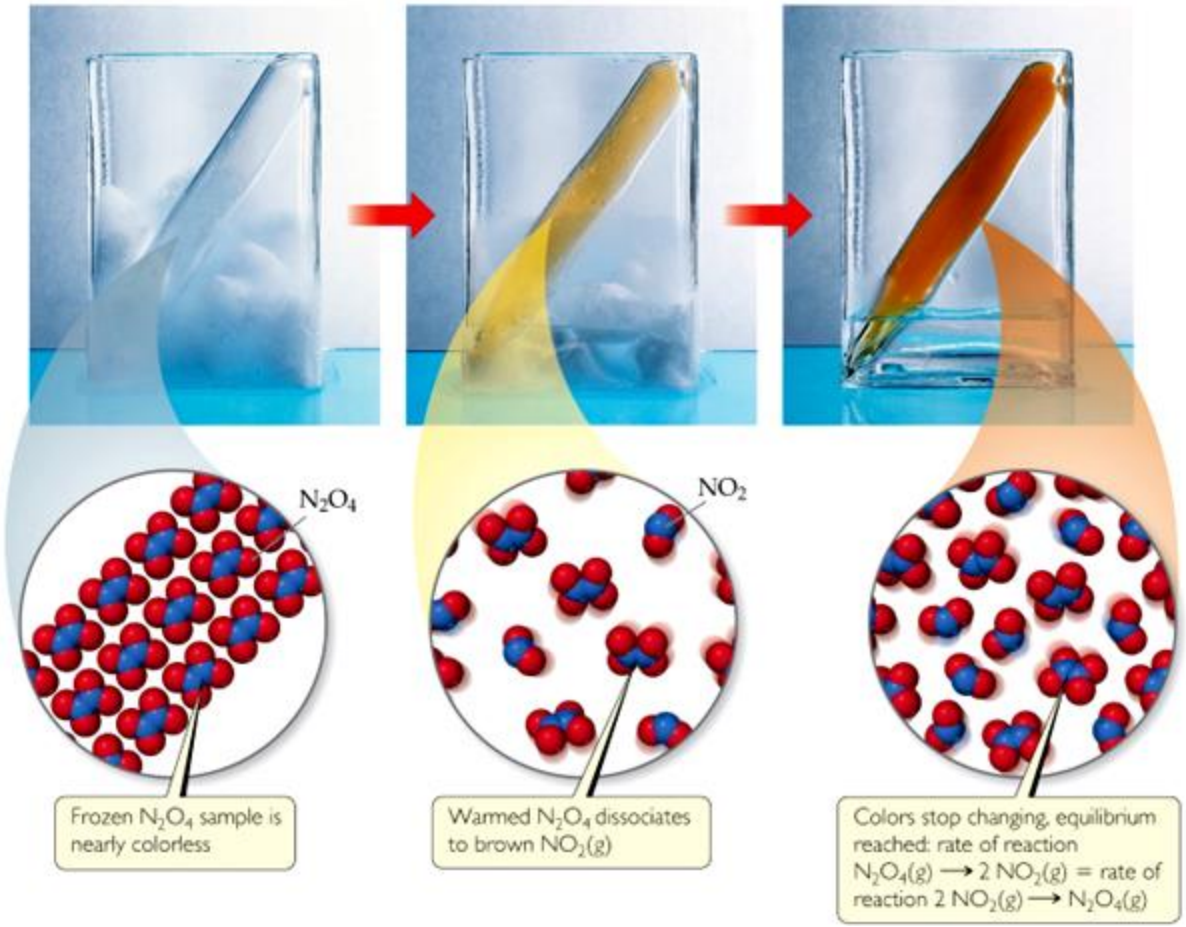
Plugging this into the expression for K_p for each substance, the relationship between K_c and K_p becomes

$$K_p = K_c\,(RT)^{\Delta n}$$

where

Δn = (moles of gaseous product) – (moles of gaseous reactant)

Equilibrium Can Be Reached from Either Direction

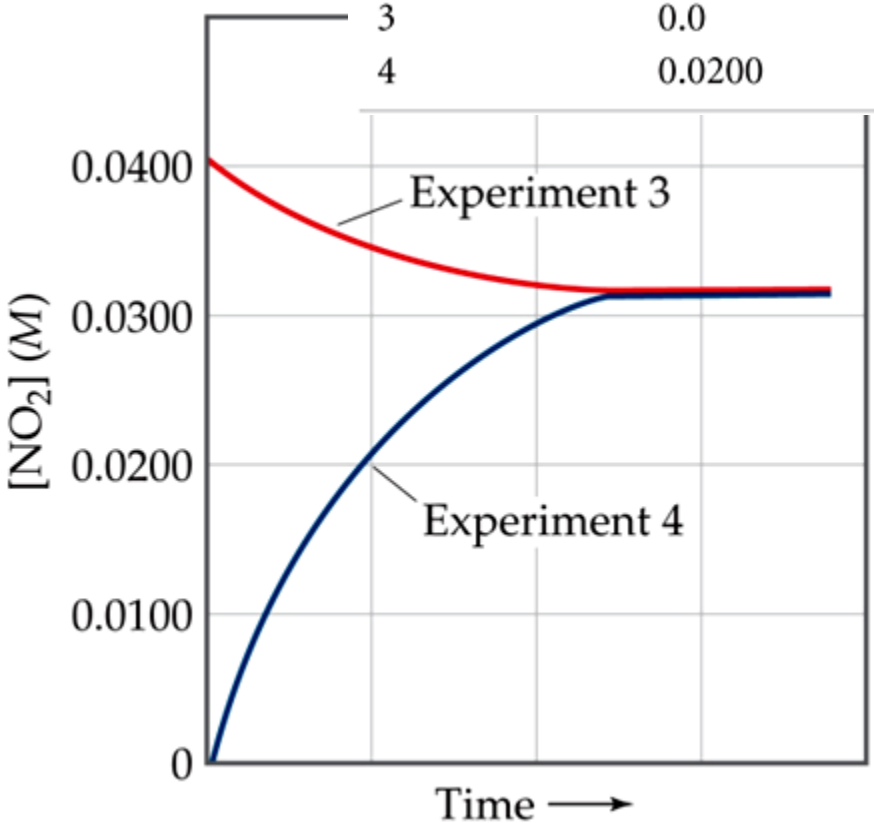


As you can see, the ratio of $[NO_2]^2$ to $[N_2O_4]$ remains constant at this temperature no matter what the initial concentrations of NO_2 and N_2O_4 are.

Equilibrium Can Be Reached from Either Direction

TABLE 15.1 • Initial and Equilibrium Concentrations of $N_2O_4(g)$ and $NO_2(g)$ at 100 °C

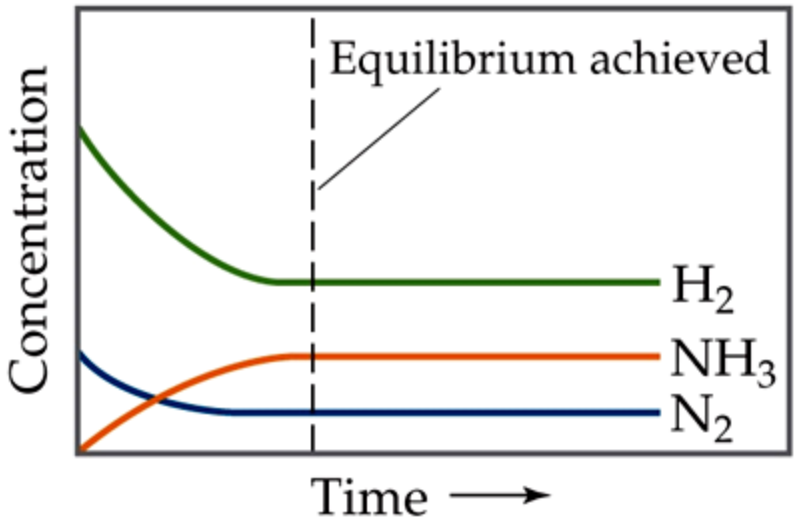
Experiment	Initial $[N_2O_4]$ (M)	Initial $[NO_2]$ (M)	Equilibrium $[N_2O_4]$ (M)	Equilibrium $[NO_2]$ (M)	K_c
1	0.0	0.0200	0.00140	0.0172	0.211
2	0.0	0.0300	0.00280	0.0243	0.211
3	0.0	0.0400	0.00452	0.0310	0.213
4	0.0200	0.0	0.00452	0.0310	0.213

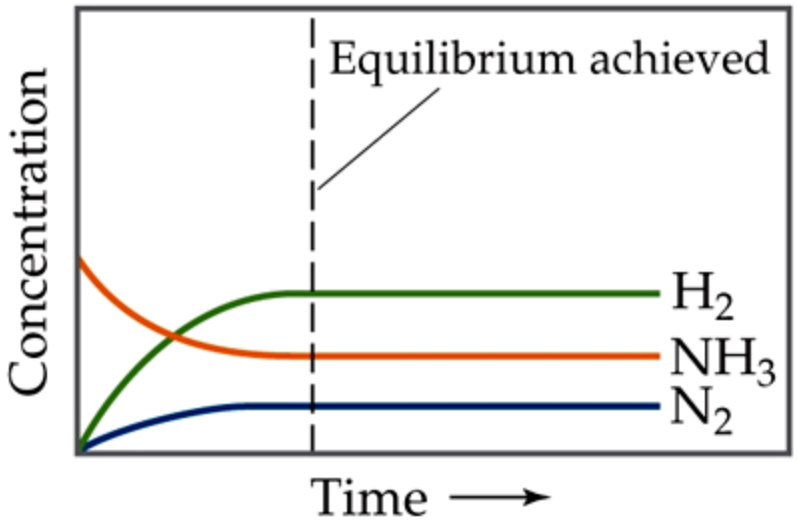


These are the data from the last two trials from the table above.

Equilibrium Can Be Reached from Either Direction

$$N_2(g) + 3H_2(g) \rightleftharpoons 3NH_3(g)$$



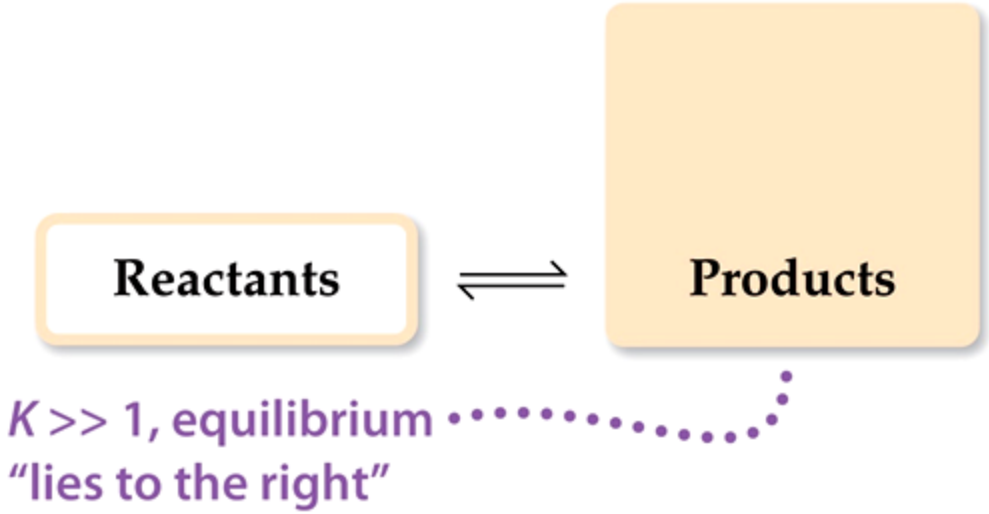


It doesn't matter whether we start with N_2 and H_2 or whether we start with NH_3, we will have the same proportions of all three substances at equilibrium.

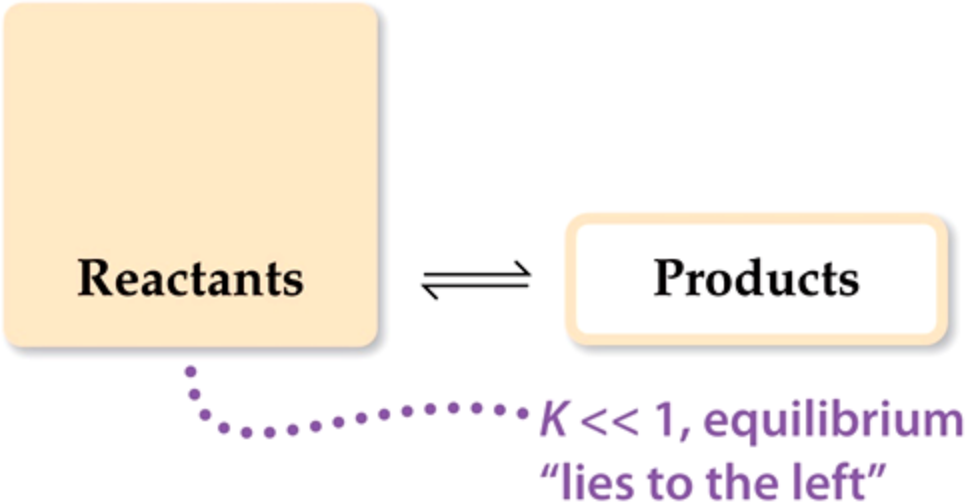
Equilibrium Constants and Units

- Equilibrium constants are reported without units.
- Equilibrium constants is related to kinetics of a reaction as well as to the thermodynamics (Chapter 19).
- Equilibrium constants derived from thermodynamic measurements are defined in terms of *activities* rather than concentrations or partial pressures.
- The *activity* of any substance in an *ideal* mixture is the ratio of the concentration or pressure of the substance either to a reference concentration (1 M) or to a reference pressure (1 atm). Therefore activities have no unit but the numerical value of the activity equal to the concentration.

What Does the Value of *K* Mean?



- If $K>>1$, the reaction is *product-favored*; product predominates at equilibrium.



- If $K<<1$, the reaction is *reactant-favored*; reactant predominates at equilibrium.

Manipulating Equilibrium Constants

The equilibrium constant of a reaction in the reverse reaction is the reciprocal 倒数 of the equilibrium constant of the forward reaction:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \quad K_c = \frac{[NO_2]^2}{[N_2O_4]} = 0.212 @ 100\ ^\circ C$$

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \quad K_c = \frac{[N_2O_4]}{[NO_2]^2} = 4.72\ @ 100\ ^\circ C$$

Whenever you are using an equilibrium constant, you should always write the associated balanced chemical reaction.

Manipulating Equilibrium Constants

The equilibrium constant of a reaction that has been multiplied by a number is the equilibrium constant raised to a power that is equal to that number:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \quad K_c = \frac{[NO_2]^2}{[N_2O_4]} = 0.212 \text{ @ } 100\ ^\circ C$$

$$2N_2O_4(g) \rightleftharpoons 4NO_2(g) \quad K_c = \frac{[NO_2]^4}{[N_2O_4]^2} = (0.212)^2 \text{ @ } 100\ ^\circ C$$

Whenever you are using an equilibrium constant, you should always write the associated balanced chemical reaction.

Manipulating Equilibrium Constants

The equilibrium constant for a net reaction made up of two or more steps is the product of the equilibrium constants for the individual steps.

$$1.\ A + B \rightleftharpoons C + D \quad K_1$$

$$2.\ C + F \rightleftharpoons G + A \quad K_2$$

$$3.\ B + F \rightleftharpoons D + G \quad \boxed{K_3 = (K_1)(K_2)}$$

Exercise Combining Equilibrium Expressions

Given the reactions

$$HF(aq) \rightleftharpoons H^+(aq) + F^-(aq) \qquad K_c = 6.8 \times 10^{-4}$$

$$H_2C_2O_4(aq) \rightleftharpoons 2\,H^+(aq) + C_2O_4^{2-}(aq) \qquad K_c = 3.8 \times 10^{-6}$$

determine the value of K_c for the reaction

$$2\,HF(aq) + C_2O_4^{2-}(aq) \rightleftharpoons 2\,F^-(aq) + H_2C_2O_4(aq)$$

Solution

If we multiply the first equation by 2 and make the corresponding change to its equilibrium constant (raising to the power 2), we get

$$2\,HF(aq) \rightleftharpoons 2\,H^+(aq) + 2\,F^-(aq) \qquad K_c = (6.8 \times 10^{-4})^2 = 4.6 \times 10^{-7}$$

Reversing the second equation and again making the corresponding change to its equilibrium constant (taking the reciprocal) gives

$$2\,H^+(aq) + C_2O_4^{2-}(aq) \rightleftharpoons H_2C_2O_4(aq) \qquad K_c = \frac{1}{3.8 \times 10^{-6}} = 2.6 \times 10^5$$

Now we have two equations that sum to give the net equation, and we can multiply the individual K_c values to get the desired equilibrium constant.

$$2\,HF(aq) \rightleftharpoons 2\,H^+(aq) + 2\,F^-(aq) \qquad K_c = 4.6 \times 10^{-7}$$

$$2\,H^+(aq) + C_2O_4^{2-}(aq) \rightleftharpoons H_2C_2O_4(aq) \qquad K_c = 2.5 \times 10^5$$

$$2\,HF(aq) + C_2O_4^{2-}(aq) \rightleftharpoons 2\,F^-(aq) + H_2C_2O_4(aq) \qquad K_c = (4.6 \times 10^{-7})(2.6 \times 10^5) = 0.12$$

Heterogeneous Equilibrium

The Concentrations of Solids and Liquids Are Essentially Constant

Both the concentrations of solids and liquids can be obtained by multiplying the density of the substance by its molar mass—and both of these are constants at constant temperature.

The Concentrations of Solids and Liquids Are Essentially Constant

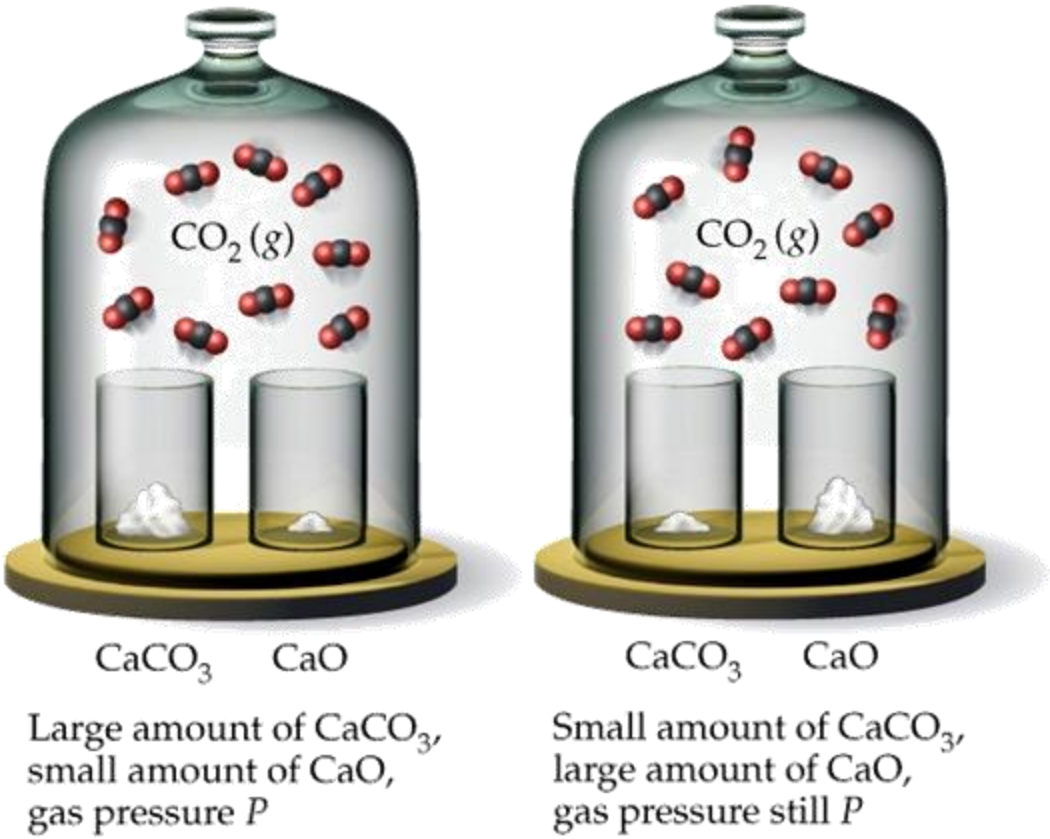
Therefore, the concentrations of solids and liquids do not appear in the equilibrium expression.

$$PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^{-}(aq)$$

$$K_c = [Pb^{2+}]\,[Cl^{-}]^2$$

The Concentrations of Solids and Liquids Are Essentially Constant

$$CaCO_3(s) \rightleftharpoons CO_2(g) + CaO(s)$$



As long as *some* $CaCO_3$ or CaO remain in the system, the amount of CO_2 above the solid will remain the same.

Exercise Analyzing a Heterogeneous Equilibrium

Each of these mixtures was placed in a closed container and allowed to stand:

(a) $CaCO_3(s)$
(b) $CaO(s)$ and $CO_2(g)$ at a pressure greater than the value of K_p
(c) $CaCO_3(s)$ and $CO_2(g)$ at a pressure greater than the value of K_p
(d) $CaCO_3(s)$ and $CaO(s)$

Determine whether or not each mixture can attain the equilibrium.

Solution

Plan For equilibrium to be achieved, it must be possible for both the forward process and the reverse process to occur.

Solve Equilibrium can be reached in all cases except (c) as long as sufficient quantities of solids are present.
(a) $CaCO_3$ simply decomposes, forming $CaO(s)$ and $CO_2(g)$ until the equilibrium pressure of CO_2 is attained. There must be enough $CaCO_3$, however, to allow the CO_2 pressure to reach equilibrium. **(b)** CO_2 continues to combine with CaO until the partial pressure of the CO_2 decreases to the equilibrium value. **(c)** There is no CaO present, so equilibrium cannot be attained because there is no way the CO_2 pressure can decrease to its equilibrium value (which would require some of the CO_2 to react with CaO). **(d)** The situation is essentially the same as in (a): $CaCO_3$ decomposes until equilibrium is attained. The presence of CaO initially makes no difference.

GIVE IT SOME THOUGHT

Write the equilibrium-constant expression for the evaporation of water, $H_2O(l) \rightleftharpoons H_2O(g)$, in terms of partial pressures.

A. $K_p = 1/[H_2O(g)]$
B. $K_p = P_{H_2O(g)}$
C. $K_p = [H_2O(g)]$
D. $K_p = 1/P_{H_2O(g)}$

GIVE IT SOME THOUGHT

Write the equilibrium-constant expression for the evaporation of water, $H_2O(l) \rightleftharpoons H_2O(g)$, in terms of partial pressures.

A. $K_p = 1/[H_2O(g)]$

B. $K_p = P_{H_2O(g)}$

C. $K_p = [H_2O(g)]$

D. $K_p = 1/P_{H_2O(g)}$

GIVE IT SOME THOUGHT

Write the equilibrium-constant expression for the reaction $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

A. $K_c = [NH_4^+][OH^-]$

B. $K_c = \dfrac{[NH_4^+][OH^-]}{[NH_3]}$

C. $K_c = \dfrac{[NH_4^+][OH^-]}{[NH_3][H_2O]}$

D. $K_c = \dfrac{[NH_3][H_2O]}{[NH_4^+][OH^-]}$

GIVE IT SOME THOUGHT

Write the equilibrium-constant expression for the reaction $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

A. $K_c = [NH_4^+][OH^-]$

B. $K_c = \dfrac{[NH_4^+][OH^-]}{[NH_3]}$

C. $K_c = \dfrac{[NH_4^+][OH^-]}{[NH_3][H_2O]}$

D. $K_c = \dfrac{[NH_3][H_2O]}{[NH_4^+][OH^-]}$

Equilibrium Calculations

An Equilibrium Problem

A closed system initially containing 1.000 x 10^{-3} *M* H_2 and 2.000 x 10^{-3} *M* I_2 at 448 °C is allowed to reach equilibrium. Analysis of the equilibrium mixture shows that the concentration of HI is 1.87 x 10^{-3} *M*. Calculate K_c at 448 °C for the reaction taking place, which is

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

An Equilibrium Problem

ICE chart, where ICE stands for Initial – Change – Equilibrium

	$[H_2]$, *M*	$[I_2]$, *M*	[HI], *M*
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change			
At equilibrium			1.87×10^{-3}

What do we know?

An Equilibrium Problem

ICE chart, where ICE stands for Initial – Change – Equilibrium

	$[H_2]$, *M*	$[I_2]$, *M*	[HI], *M*
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change			$\mathbf{+1.87 \times 10^{-3}}$
At equilibrium			1.87×10^{-3}

[HI] Increases by 1.87×10^{-3} *M*

An Equilibrium Problem

ICE chart, where ICE stands for Initial – Change – Equilibrium

	$[H_2]$, *M*	$[I_2]$, *M*	[HI], *M*
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change	$\mathbf{-9.35 \times 10^{-4}}$	$\mathbf{-9.35 \times 10^{-4}}$	$+1.87 \times 10^{-3}$
At equilibrium			1.87×10^{-3}

Stoichiometry tells us $[H_2]$ and $[I_2]$ decrease by half as much.

An Equilibrium Problem

ICE chart, where ICE stands for Initial – Change – Equilibrium

	$[H_2]$, *M*	$[I_2]$, *M*	[HI], *M*
Initially	1.000×10^{-3}	2.000×10^{-3}	0
Change	-9.35×10^{-4}	-9.35×10^{-4}	$+1.87 \times 10^{-3}$
At equilibrium	$\mathbf{6.5 \times 10^{-5}}$	$\mathbf{1.065 \times 10^{-3}}$	1.87×10^{-3}

We can now calculate the equilibrium concentrations of all three compounds

An Equilibrium Problem

And, therefore, the equilibrium constant:

$$K_c = \frac{[HI]^2}{[H_2]\,[I_2]}$$

$$= \frac{(1.87 \times 10^{-3})^2}{(6.5 \times 10^{-5})(1.065 \times 10^{-3})}$$

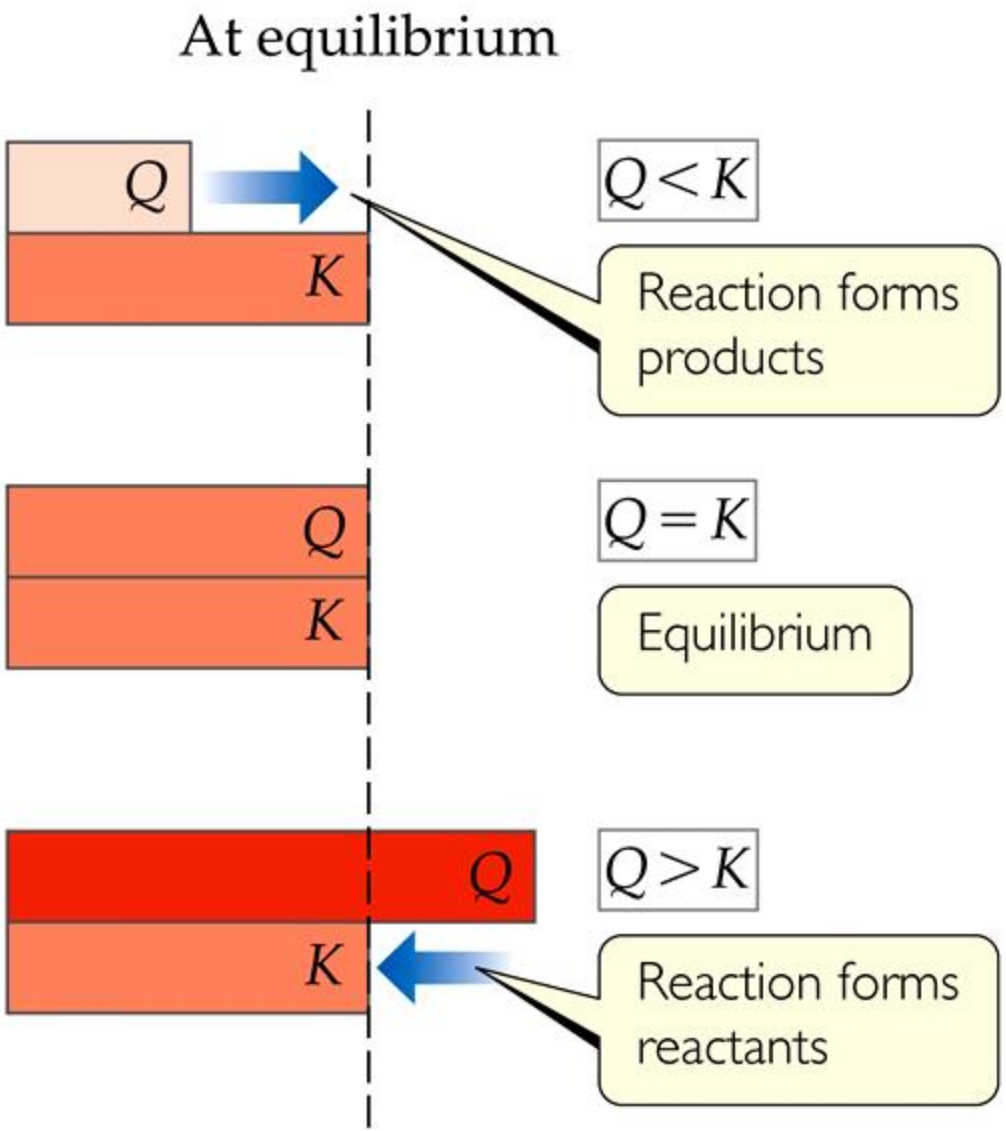
$$= 51$$

Two significant figures

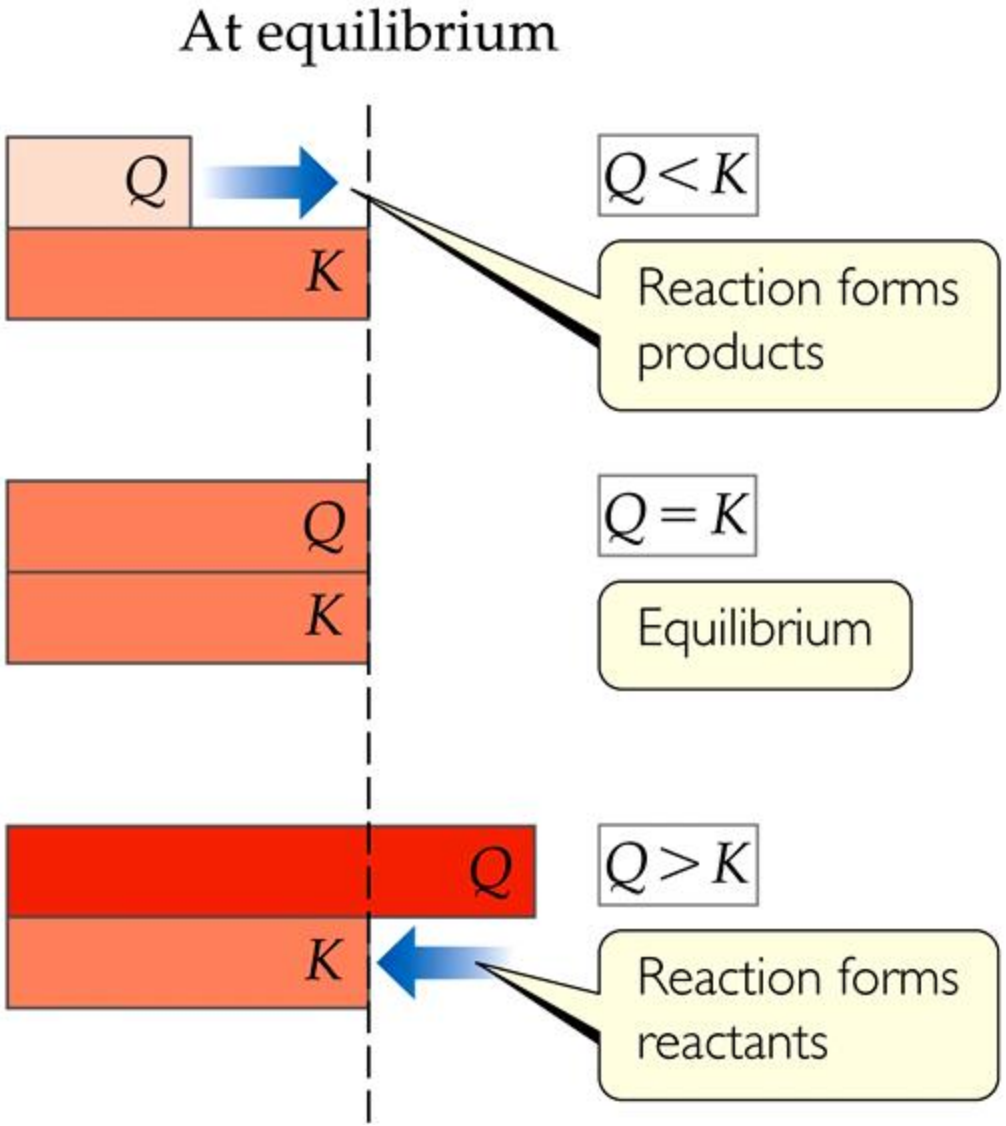
The Reaction Quotient (*Q*) 商

- *Q* gives the same ratio the equilibrium expression gives, **but for a system that is *not* at equilibrium**.
- To calculate *Q*, one substitutes the initial concentrations on reactants and products into the equilibrium expression.

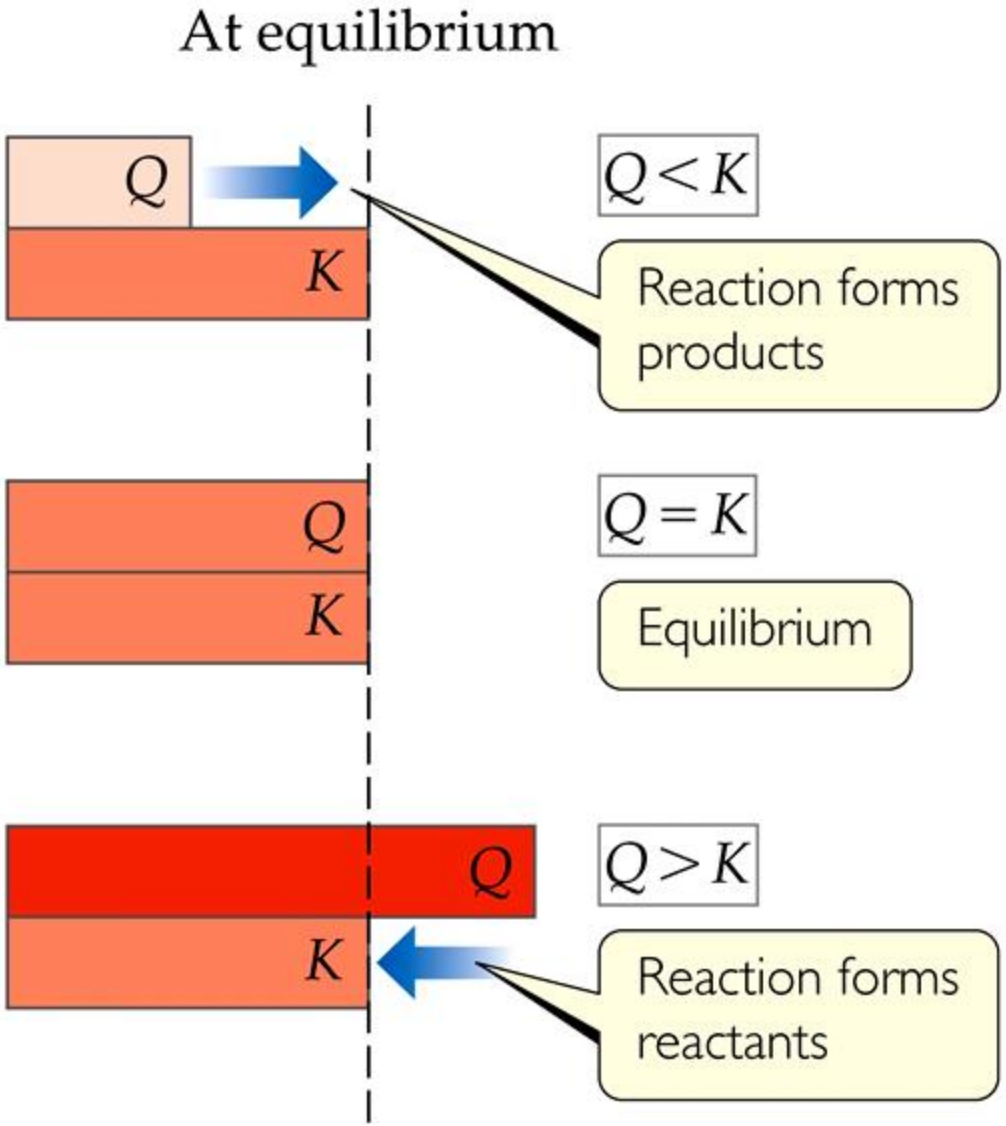
If $Q = K$, the system is at equilibrium.



If $Q > K$, there is too much product, and the equilibrium shifts to the left.



If $Q < K$, there is too much reactant, and the equilibrium shifts to the right.



Exercise Predicting the Direction of Approach to Equilibrium

At 448 °C the equilibrium constant K_c for the reaction is 50.5.

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

Predict in which direction the reaction proceeds to reach equilibrium if we start with 2.0×10^{-2} mol of HI, 1.0×10^{-2} mol of H_2, and 3.0×10^{-2} of I_2 in a 2.00-L container.

Solution

The initial concentrations are

$$[HI] = 2.0 \times 10^{-2}\ \text{mol}/2.00\ \text{L} = 1.0 \times 10^{-2}\ M$$
$$[H_2] = 1.0 \times 10^{-2}\ \text{mol}/2.00\ \text{L} = 5.0 \times 10^{-3}\ M$$
$$[I_2] = 3.0 \times 10^{-2}\ \text{mol}/2.00\ \text{L} = 1.5 \times 10^{-2}\ M$$

The reaction quotient is therefore

$$Q_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(1.0 \times 10^{-2})^2}{(5.0 \times 10^{-3})(1.5 \times 10^{-2})} = 1.3$$

Because $Q_c < K_c$, the concentration of HI must increase and the concentrations of H_2 and I_2 must decrease to reach equilibrium; the reaction as written proceeds left to right to attain equilibrium.

Exercise Calculating Equilibrium Concentration from Initial Concentrations

A 1.000-L flask is filled with 1.000 mol of $H_2(g)$ and 2.000 mol of $I_2(g)$ at 447 °C. The value of the equilibrium constant K_c for the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

at 448 °C is 50.5.What are the equilibrium concentrations of H_2, I_2, and HI in moles per liter?

Solution

First, we note the initial concentrations of H_2 and I_2:

$$[H_2] = 1.000\ M \text{ and } [I_2] = 2.000\ M$$

Second, we construct a table in which we tabulate the initial concentrations:

	$H_2(g)$ +	$I_2(g)$ $\rightleftharpoons$	$2\,HI(g)$
Initial concentration (M)	1.000	2.000	0
Change in concentration (M)			
Equilibrium concentration (M)			

Exercise Calculating Equilibrium Concentration from Initial Concentrations

Third, we use the stoichiometry of the reaction to determine the changes in concentration that occur as the reaction proceeds to equilibrium. The H_2 and I_2 concentrations will decrease as equilibrium is established and that of HI will increase. Let's represent the change in concentration of H_2 by *x*. The balanced chemical equation tells us the relationship between the changes in the concentrations of the three gases. For each *x* mol of H_2 that reacts, *x* mol of I_2 are consumed and 2*x* mol of HI are produced:

	$H_2(g)$ +	$I_2(g)$ $\rightleftharpoons$	$2\ HI(g)$
Initial concentration (M)	1.000	2.000	0
Change in concentration (M)	$-x$	$-x$	$+2x$
Equilibrium concentration (M)			

Fourth, we use initial concentrations and changes in concentrations, as dictated by stoichiometry, to express the equilibrium concentrations. With all our entries, our table now looks like this:

	$H_2(g)$ +	$I_2(g)$ $\rightleftharpoons$	$2\ HI(g)$
Initial concentration (M)	1.000	2.000	0
Change in concentration (M)	$-x$	$-x$	$+2x$
Equilibrium concentration (M)	$1.000 - x$	$2.000 - x$	$2x$

Exercise Calculating Equilibrium Concentration from Initial Concentrations

Fifth, we substitute the equilibrium concentrations into the equilibrium-constant expression and solve for *x*:

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2x)^2}{(1.000 - x)(2.000 - x)} = 50.5$$

If you have an equation-solving calculator, you can solve this equation directly for *x*. If not, expand this expression to obtain a quadratic equation in *x*:

$$4x^2 = 50.5(x^2 - 3.000x + 2.000)$$

$$46.5x^2 - 151.5x + 101.0 = 0$$

Solving the quadratic equation (Appendix A.3) leads to two solutions for *x*:

$$x = \frac{-(-151.5) \pm \sqrt{(-151.5)^2 - 4(46.5)(101.0)}}{2(46.5)} = 2.323 \text{ or } 0.935$$

When we substitute *x* = 2.323 into the expressions for the equilibrium concentrations, we find *negative* concentrations of H_2 and I_2. Because a negative concentration is not chemically meaningful, we reject this solution. We then use *x* = 0.935 to find the equilibrium concentrations:

$$[H_2] = 1.000 - x = 0.065\ M$$

$$[I_2] = 2.000 - x = 1.065\ M$$

$$[HI] = 2x = 1.87\ M$$

Le Châtelier's Principle

Henri-Louis Le Châtelier (1850–1936) 勒夏特列

Le Châtelier's Principle

"If a system at equilibrium is disturbed by a change in temperature, pressure, or the concentration of one of the components, the system will shift its equilibrium position so as to counteract the effect of the disturbance."

Le Châtelier's Principle

Effects of Concentration and Temperature Change

Exothermic

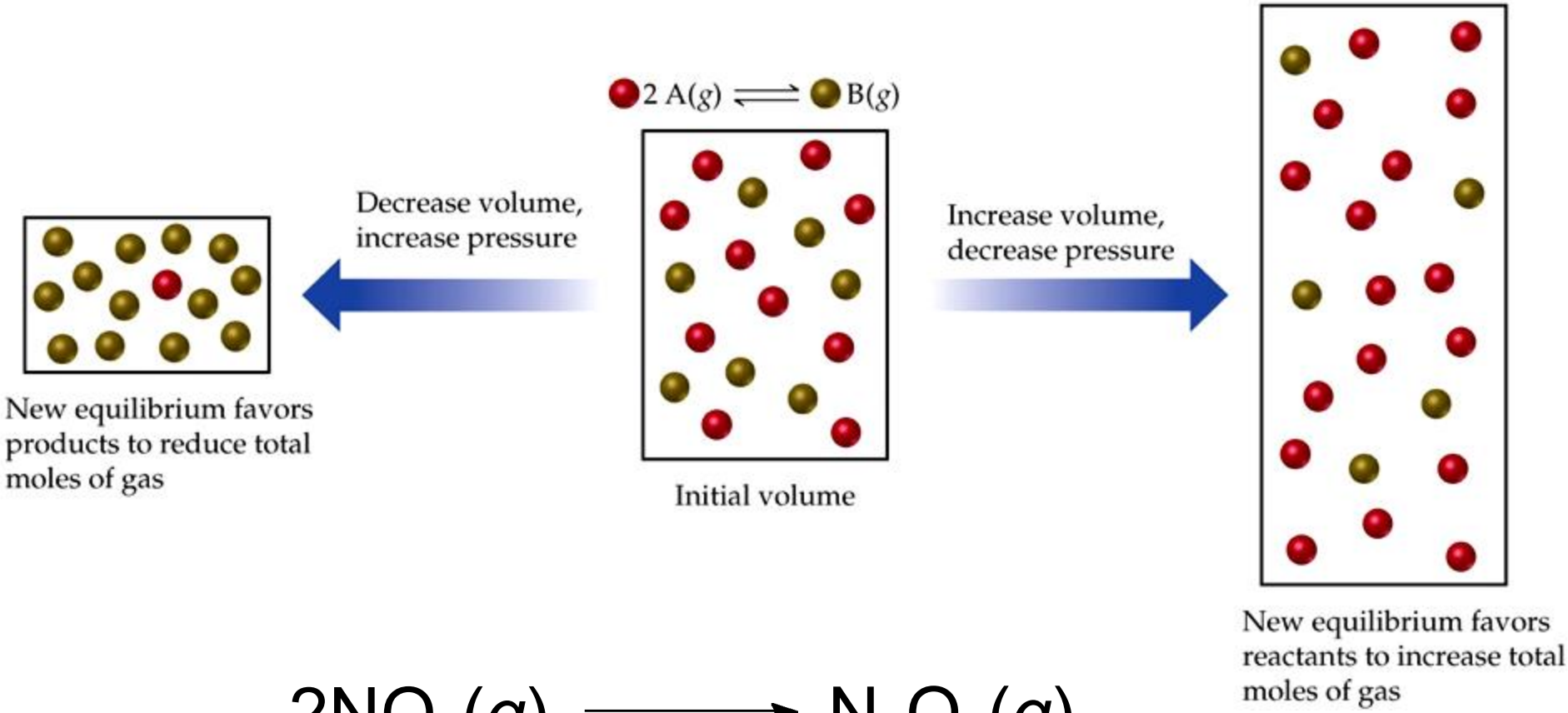
$$CoCl_4^{2-}(aq) + 6H_2O(l) \underset{\Delta H>0}{\overset{\Delta H<0}{\rightleftharpoons}} Co(H_2O)_6^{2+}(aq) + 4Cl^-(aq)$$

Blue **Pink**

Endothermic

Le Châtelier's Principle

Effects of Volume and Pressure Changes



$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

Le Châtelier's Principle

System	Change	Result
$CO_2 + H_2 \rightarrow H_2O(g) + CO$	A drying agent is added to absorb H_2O	**Shift to the right**. Continuous removal of a product will force any reaction to the right
$H_2(g) + I_2(g) \rightarrow 2HI(g)$	Some nitrogen gas is added	**No change**; N_2 is not a component of this reaction system.
$NaCl(s) + H_2SO_4(l) \rightarrow Na_2SO_4(s) + HCl(g)$	Reaction is carried out in an open container	Because HCl is a gas that can escape from the system, the reaction is forced to the **right**. This is the basis for the commercial production of hydrochloric acid.
$H_2O(l) \rightarrow H_2O(g)$	Water evaporates from an open container	Continuous removal of water vapor forces the reaction to the **right**, so equilibrium is never achieved.
$HCN(aq) \rightarrow H^+(aq) + CN^-(aq)$	The solution is diluted	Shift to **right**; the product $[H^+][CN^-]$ diminishes more rapidly than does [HCN].
$AgCl(s) \rightarrow Ag^+(aq) + Cl^-(aq)$	Some NaCl is added to the solution	Shift to **left** due to increase in Cl^- concentration. This is known as the common ion effect on solubility.
$N_2 + 3\ H_2 \rightarrow 2\ NH_3$	A catalyst is added to speed up this reaction	**No change**. Catalysts affect only the rate of a reaction; the have no effect at all on the composition of the equilibrium state.

GIVE IT SOME THOUGHT

Use Le Châtelier's principle to explain why the equilibrium vapor pressure of a liquid increases with increasing temperature.

A. Evaporation is an exothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the right and increases the equilibrium vapor pressure.

B. Evaporation is an endothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the right and increases the equilibrium vapor pressure.

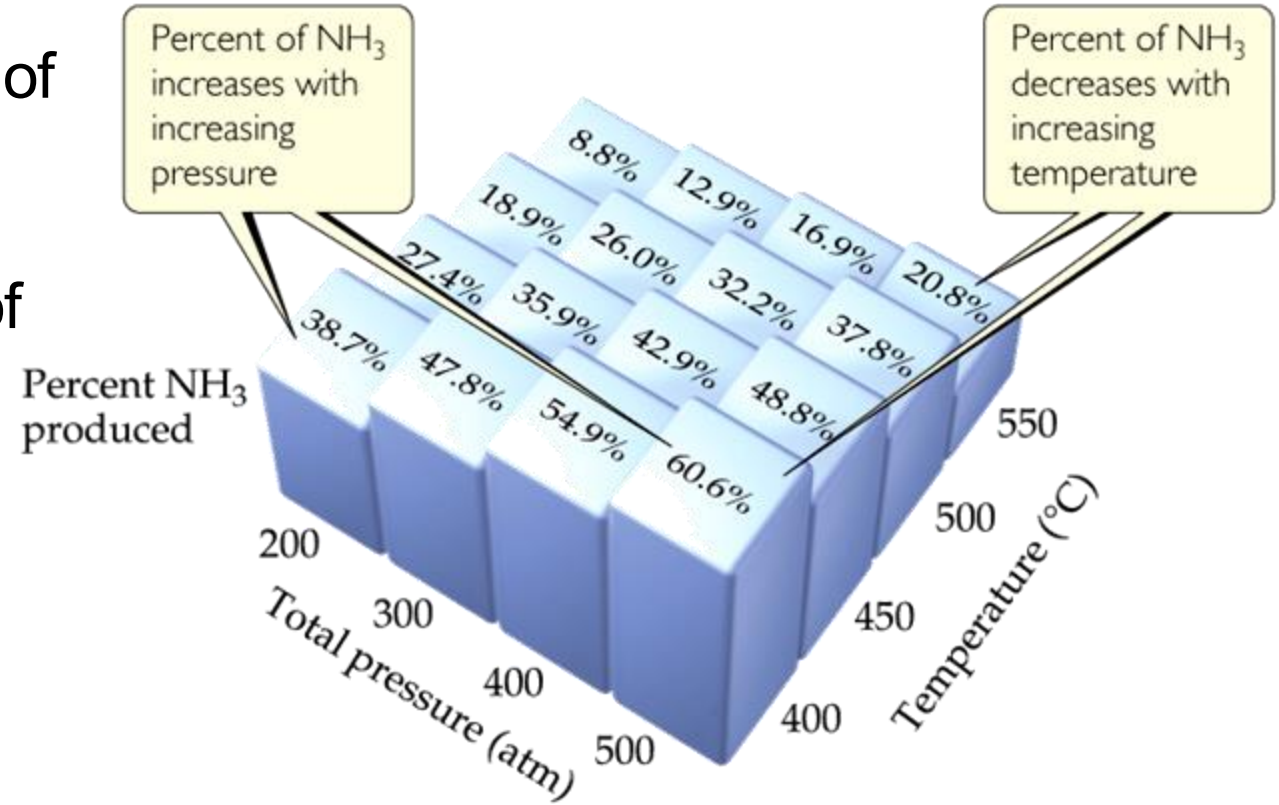
C. Evaporation is an exothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the left and increases the equilibrium vapor pressure.

D. Evaporation is an endothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the left and increases the equilibrium vapor pressure.

The Haber Process

$$N_2(g) + 3\ H_2(g) \longrightarrow 2\ NH_3(g) \qquad \Delta H = -92\ \text{kJ/mol}$$

The transformation of nitrogen and hydrogen into ammonia (NH_3) is of tremendous significance in agriculture, where ammonia-based fertilizers are of utmost importance.



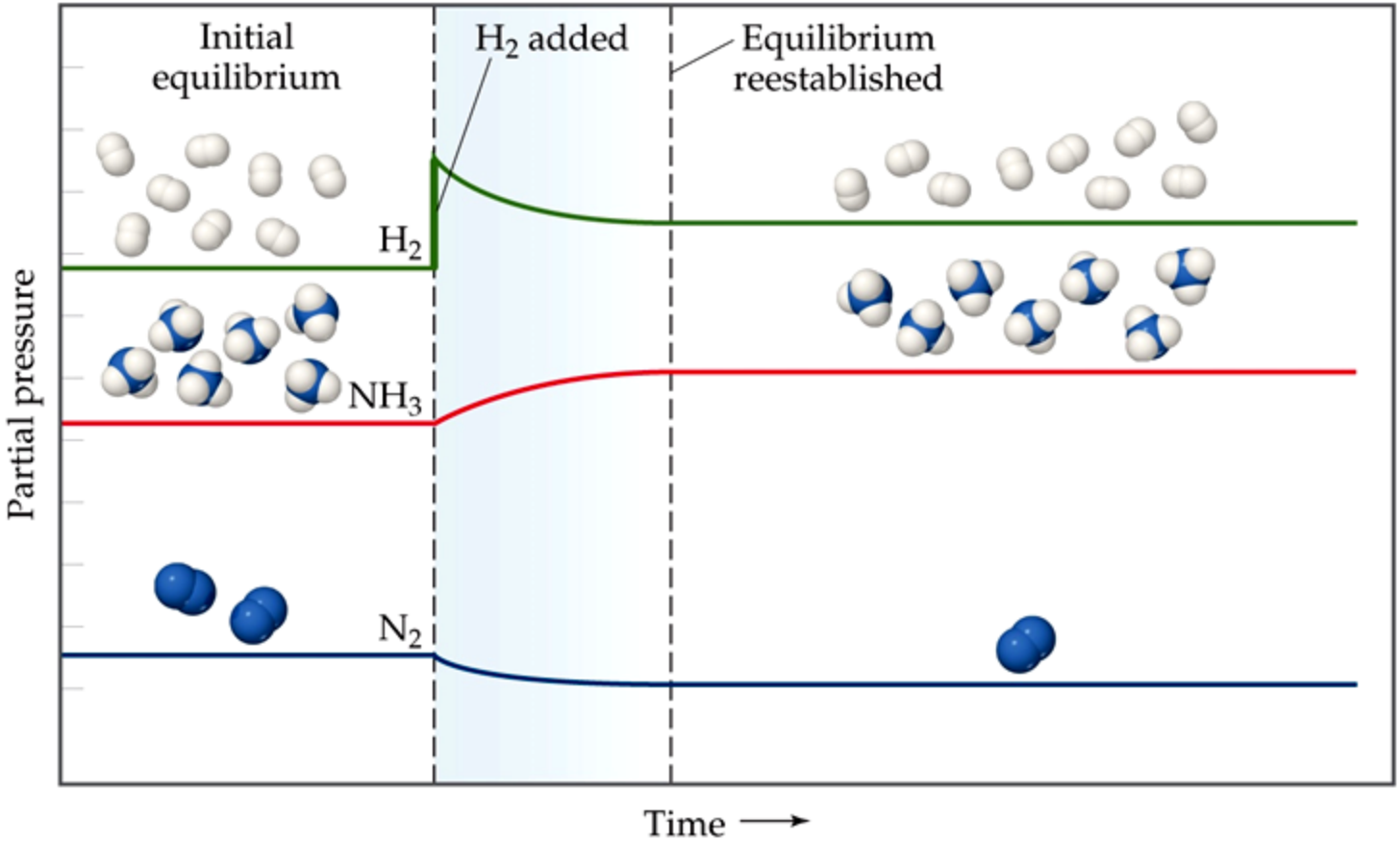
Changes in Temperature

$N_2(g) + 3\ H_2(g) \longrightarrow 2\ NH_3(g)$ $\Delta H = -92$ kJ/mol

Temperature (°C)	K_p
300	4.34×10^{-3}
400	1.64×10^{-4}
450	4.51×10^{-5}
500	1.45×10^{-5}
550	5.38×10^{-6}
600	2.25×10^{-6}

The Haber Process

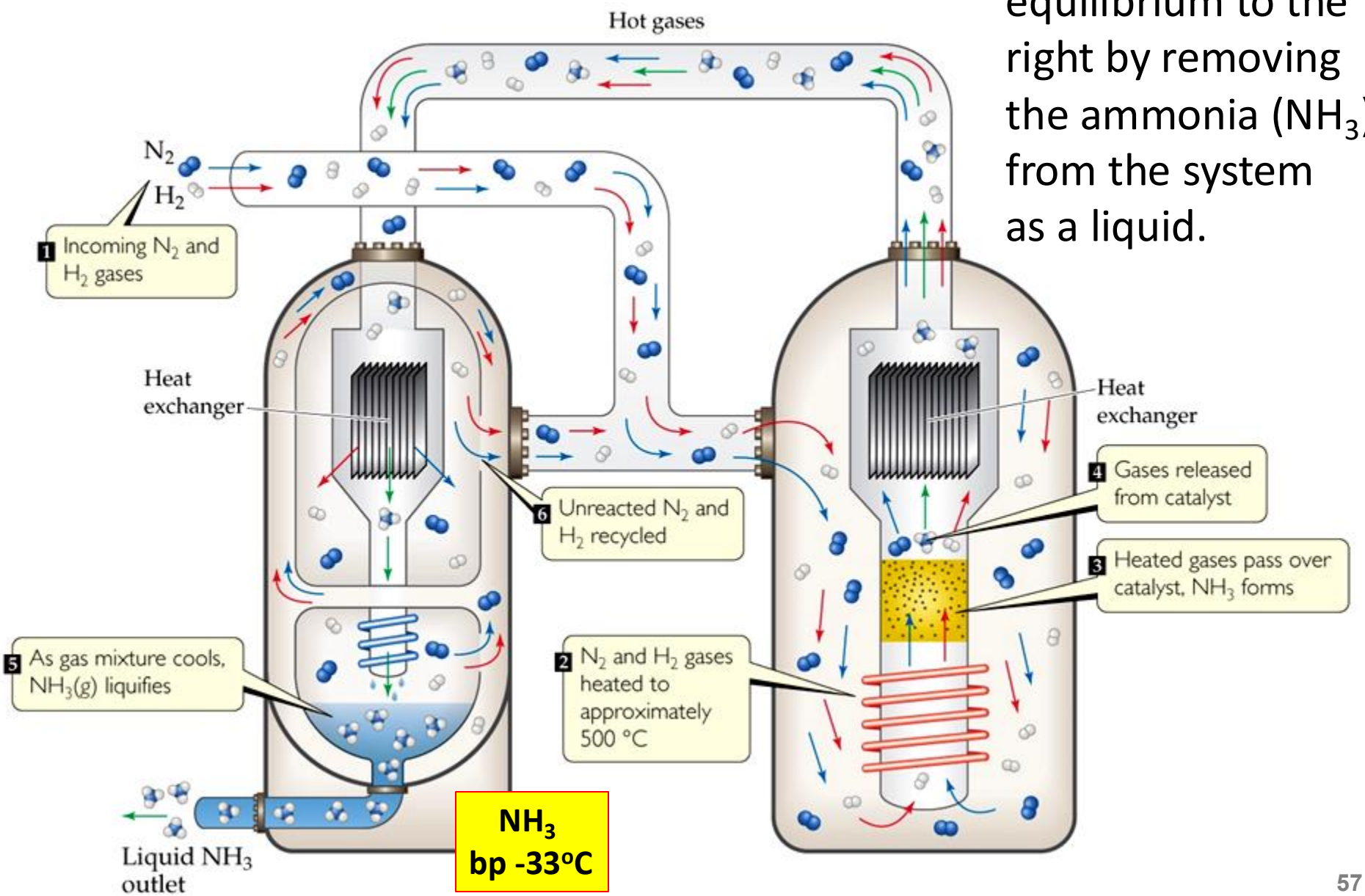
$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$$



If H_2 is added to the system, N_2 will be consumed and the two reagents will form more NH_3.

The Haber Process

This apparatus helps push the equilibrium to the right by removing the ammonia (NH_3) from the system as a liquid.

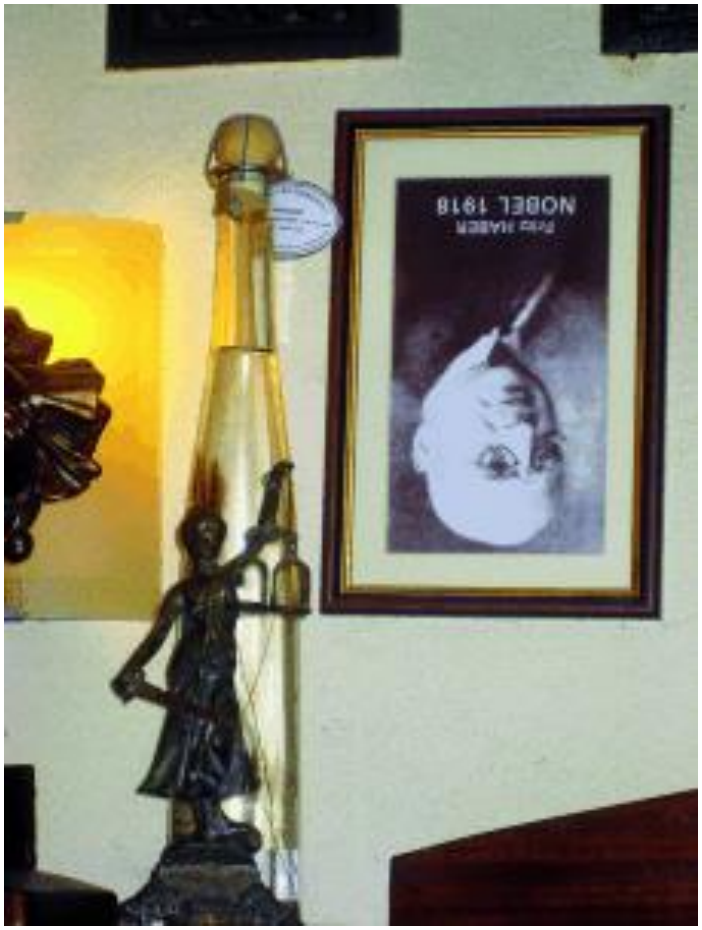


Changes in Temperature

TABLE 15.2 • Variation in K_p with Temperature for $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$

Temperature (°C)	K_p
300	4.34×10^{-3}
400	1.64×10^{-4}
450	4.51×10^{-5}
500	1.45×10^{-5}
550	5.38×10^{-6}
600	2.25×10^{-6}

Fritz Haber: The Damned Scientist



Fritz Haber (1868 – 1934)

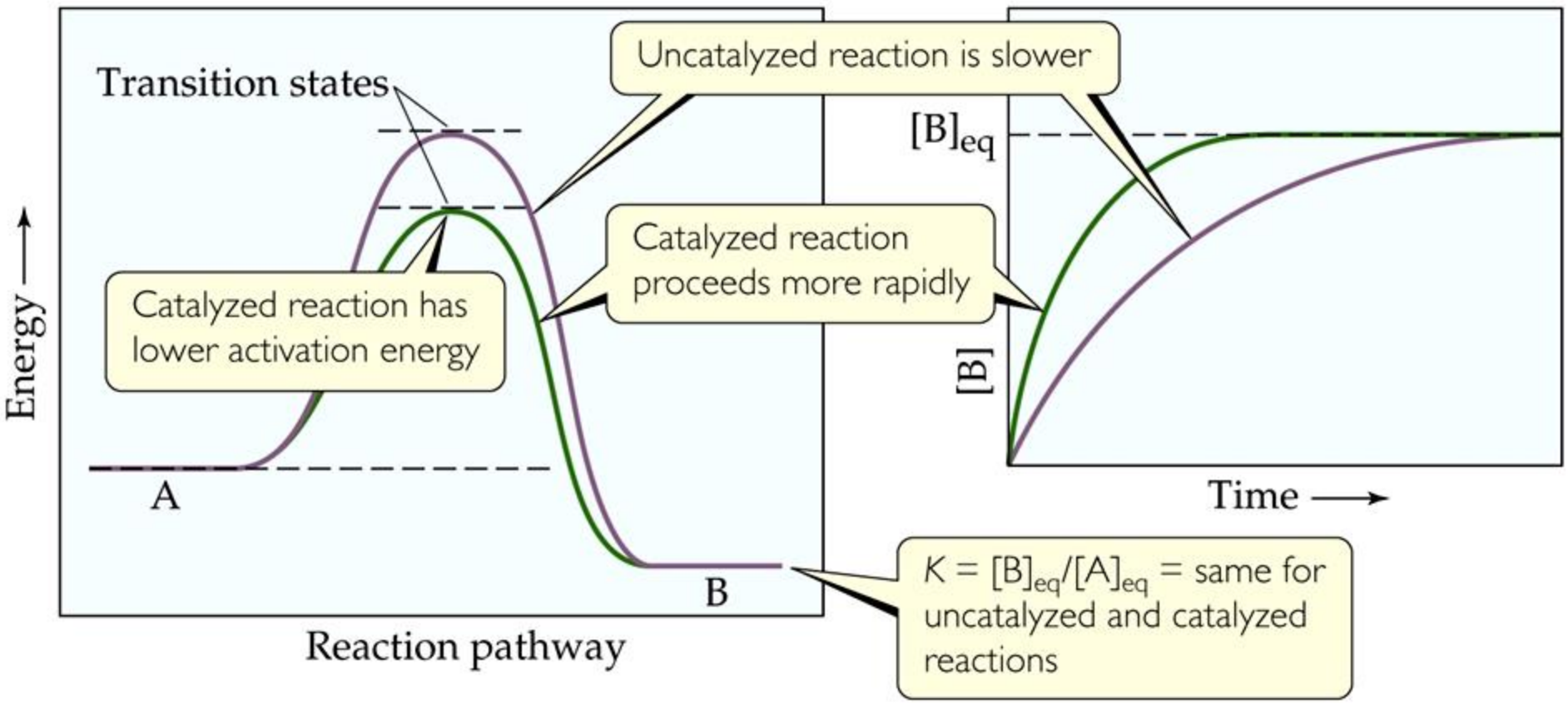
Angew. Chem. Int. Ed. **2011**, *50*, 10050 – 10062

http://onlinelibrary.wiley.com/doi/10.1002/anie.201105425/full#fig2

Catalysts

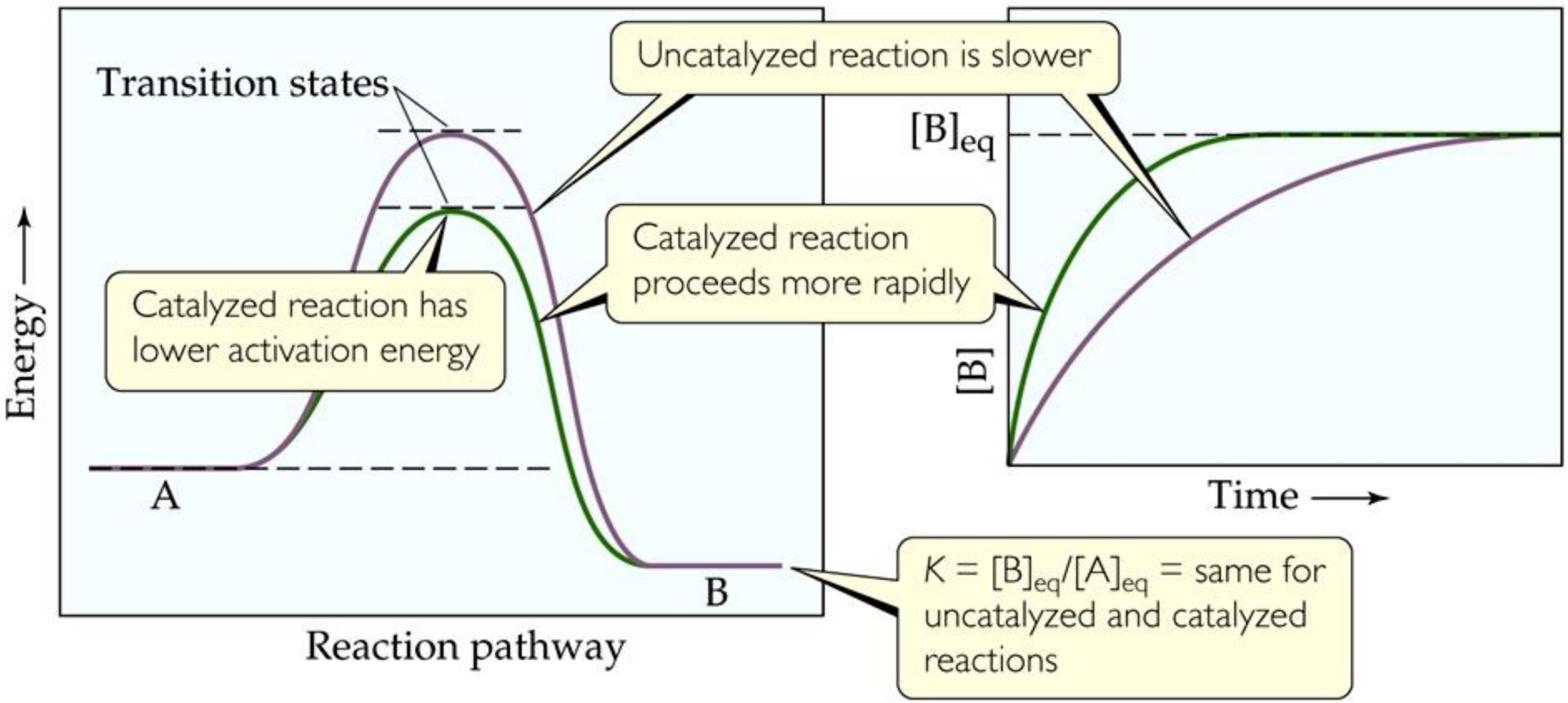
Catalysts

Catalysts increase the rate of both the forward *and* reverse reactions.



Catalysts

When one uses a catalyst, equilibrium is achieved faster, but the equilibrium composition remains unaltered.



At equilibrium, the rate of the forward reaction is _____ the rate of the reverse reaction.

a. equal to
b. slower than
c. faster than
d. the reverse of

The reaction quotient Q is usually represented by

a. [reactants] / [products].
b. [products] / [reactants].
c. [reactants] × [products].
d. [reactants] + [products].

If the value of the equilibrium constant is large, then _____ will mostly be present at equilibrium.

a. reactants
b. products
c. catalysts
d. shrapnel

If the value of the equilibrium constant is small, then _____ will mostly be present at equilibrium.

a. reactants
b. products
c. catalysts
d. shrapnel

Q = the reaction quotient
K = the equilibrium constant
At equilibrium, which is true?

a. $Q > K$
b. $Q < K$
c. $Q = K$
d. $Q^2 = K$

Equilibrium constants typically have units of

a. M.
b. M^2.
c. M^{1-}.
d. None of the above

Reaction quotients for heterogeneous equilibria do not include concentrations of

a. pure liquids.
b. pure solids.
c. Both of the above
d. Neither of the above

$HA \leftrightarrow H^+ + A^-$

$[HA] = 1.65 \times 10^{-2}$ M and

$[H^+] = [A^-] = 5.44 \times 10^{-4}$ M at equilibrium. K_c = _____.

a. 1.79×10^{-2}

b. 1.79×10^{-3}

c. 1.79×10^{-4}

d. 1.79×10^{-5}

$K_P = K_C$ when

a. the reaction is at equilibrium.
b. the reaction is exothermic.
c. all of the gases present are at the same pressure.
d. the number of moles of gas on both sides of the balanced equation is the same.

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and H_2 is added, the concentration of CO at equilibrium will

a. increase.
b. decrease.
c. remain unchanged.
d. disappear.

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and H_2O is added, the equilibrium concentration of CO

a. will increase.
b. will decrease.
c. will remain unchanged.
d. will disappear.

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and CO_2 is removed, the [CO] at equilibrium will

a. increase.

b. decrease.

c. remain unchanged.

d. disappear.

$CO_2 + H_2 \leftrightarrow CO + H_2O$

Increasing the temperature of this endothermic reaction will _____ [CO] at equilibrium.

a. increase
b. decrease
c. not change
d. eradicate

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and the container is compressed, the amount of CO will

a. increase.

b. decrease.

c. remain unchanged.

d. vanish.

$CO_2 + H_2 \leftrightarrow CO + H_2O$

Adding a catalyst to this reaction will cause the [CO] at equilibrium to

a. increase.
b. decrease.
c. remain unchanged.
d. cease to exist.